

Open Computational Tools for Financial Literacy in High-Inflation Economies: The Case of *hacecuentas.com*

Working Paper · June 2026

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Abstract

Financial literacy is low worldwide, and the cost of poor financial decisions is amplified in high-inflation economies, where nominal values lose meaning within months and where indexation rules, tax brackets and wage floors are revised continually. This paper is a descriptive case study of *hacecuentas.com*, a free, open-access platform of 4,124 computational calculators (Spanish, English and Portuguese; localized for Argentina, Spain, Mexico, Chile and Colombia, spanning 7,314 indexable pages), whose largest single domain is personal finance and taxation. We describe (i) the corpus and its category distribution; (ii) a data-freshness architecture that keeps high-stakes (“Your-Money-or-Your-Life”, YMYL) calculators aligned with official sources — the central-bank rent index, the tax authority’s simplified-regime brackets, and statutory wage and social-security parameters; and (iii) interface choices that lower the cognitive cost of financial computation. We argue that maintained, source-anchored web calculators are an under-studied complement to formal financial education, especially where macroeconomic volatility renders static teaching material obsolete within a quarter. In such settings the binding constraint on usefulness is not building the tool but keeping it correct. We close with the platform’s limitations and an agenda for measuring learning and decision outcomes.

Keywords: financial literacy; financial education; inflation; indexation; Argentina; web-based tools; open educational resources; YMYL; just-in-time learning.

JEL classification: D14 (Household Saving; Personal Finance); G53 (Financial Literacy); I22 (Educational Finance); E31 (Price Level; Inflation).

1. Introduction

Financial decisions rest on arithmetic that many adults find difficult to perform reliably: how much to set aside each month, whether a loan instalment is affordable once inflation is accounted for, how this year’s salary compares with last year’s in real terms, or which simplified-tax bracket a freelancer now falls into. Three decades of survey evidence show that financial literacy is low across both developing and advanced economies, and that the gap correlates with worse borrowing, saving and retirement-planning outcomes (Lusardi & Mitchell, 2014; Klapper, Lusardi & van Oudheusden, 2015; Hastings, Madrian & Skimmyhorn, 2013). The problem is not purely conceptual. Even individuals who grasp a principle often fail at the second step: executing the computation that would apply it to their own numbers.

This paper examines one practical and surprisingly under-studied response to that *execution* gap: free, open-access web **calculators**. We present a descriptive case study of *hacecuentas.com*, a Spanish-first platform launched in 2024 that now hosts 4,124 calculator definitions across personal finance, taxation, health, education, cooking, construction and twenty other domains. Rather than teaching a concept in the abstract, each tool lets a user enter their own figures and obtain a deterministic, explained result at the moment of decision.

The Argentine setting makes the case unusually sharp. Under chronic high inflation, nominal figures decay quickly, indexation regimes proliferate (rents indexed to a central-bank series; deposits and loans denominated in inflation-linked units), and official parameters — tax brackets, social-security floors, statutory wage adjustments — are revised monthly or quarterly. A calculator that was exactly correct last

quarter can silently mislead this one. We therefore give particular attention to the **maintenance** problem: how a corpus of thousands of money-related tools is kept aligned with authoritative sources. Our central claim is that in a volatile macroeconomic environment the binding constraint on a financial tool's usefulness and trustworthiness is not its initial construction but its continued correctness over time.

The paper makes four contributions. First, it provides a transparent, quantitative description of a large open calculator corpus and its category distribution (Section 3). Second, and most importantly, it documents a **data-freshness architecture** for high-stakes calculators — single-source-of-truth parameter modules, automated drift and staleness detection, and source-anchored computation — that we offer as a reusable pattern for any maintainer of YMYL tools (Section 4). Third, it frames calculators as *executable rules of thumb*, connecting them to evidence that simple, actionable heuristics outperform comprehensive instruction (Sections 2.3 and 6.1). Fourth, it sets out an agenda for measuring whether such tools actually improve knowledge and decisions (Section 7). We are explicit about scope: this is a descriptive and design paper, not a causal impact evaluation, and Section 6.3 treats that limitation directly.

2. Background

2.1 The financial-literacy gap

Large cross-country surveys converge on a stark figure: only about one in three adults worldwide can answer a small set of questions on interest compounding, inflation and risk diversification (Klapper et al., 2015). Latin America scores below that global average, and Argentina sits within the lower band of the region. Financial literacy is not a cosmetic competence: it predicts retirement planning, wealth accumulation, debt management and the avoidance of high-cost borrowing (Lusardi & Mitchell, 2014; Lusardi, 2019).

The natural policy response is financial education, and a recent meta-analysis of 76 randomized experiments finds that it does work — financial education has economically meaningful, statistically significant effects on both financial knowledge and downstream behaviors (Kaiser, Lusardi, Menkhoff & Urban, 2022). But the same literature reports two practical caveats: effects decay as the time between instruction and decision grows, and intensive curricula are costly to deliver at scale. Both caveats point toward lightweight, *just-in-time* interventions that meet the user at the moment a calculation is actually needed.

2.2 Why high inflation raises the stakes

Argentina is among the world's most persistently inflationary economies, and the reliability of its price statistics has itself been a subject of formal study (Cavallo, 2013; Cavallo & Rigobon, 2016). High inflation degrades financial decision-making through at least three channels. First, *money illusion*: people reason in nominal terms and systematically misjudge real changes in wages, prices and debts. Second, *indexation complexity*: contracts are increasingly written against indices rather than fixed amounts — residential rents tied to a published central-bank locative-contract index, and savings and loans denominated in inflation-linked units — so that even reading one's own contract requires a computation. Third, *parameter churn*: the thresholds that determine taxes, benefits and statutory bonuses are updated several times a year, so the knowledge needed to act correctly has a short half-life.

The combined effect is that static educational material — a printed worksheet, a fixed table of brackets, a blog post with last year's numbers — becomes not merely outdated but actively misleading within a quarter. This is the environment in which a maintained, source-anchored calculator has the most to offer.

2.3 Calculators as a complement to education

A calculator is not a substitute for understanding, but it is an efficient complement to it. The clearest empirical support comes from Drexler, Fischer and Schoar (2014), who show in a randomized trial with microentrepreneurs that **rule-of-thumb** training — simple, actionable heuristics — improved financial

practices more than a conventional, comprehensive accounting course. A well-designed calculator operationalizes exactly this insight: it encodes the correct rule once, hides the bookkeeping, and delivers the answer in the user's own terms precisely when a decision is on the table. Framed as open educational resources, such tools are free to use, require no account, and — when built on a static, edge-served architecture — load on low-end devices and weak connections, which matters for equity (Section 6.2).

3. The platform: corpus and architecture

3.1 Overview and scale

hacecuentas.com is a free, advertising-light web platform launched in 2024 and operated from Buenos Aires. As of June 2026 it comprises **4,124 calculator definitions**: 2,847 in the primary Argentine-Spanish corpus, 686 in English, 219 in Portuguese, 87 in locale-neutral Spanish, and country-localized variants for Chile (99), Colombia (101) and Mexico (85). Together these resolve to **7,314 indexable pages**.

Architecturally, each calculator is the pairing of two artifacts: a **structured content record** (a JSON document holding the prompt, the input fields and their units, a plain-language explanation, a frequently-asked-questions block of at least seven entries, and a list of authoritative sources) and a **formula module** (a TypeScript function that computes the result deterministically, with no server round-trip and no opaque model). The site is rendered as static pages and served from an edge content-delivery network, so computation happens on the user's device and pages remain fast and cheap to serve at scale.

3.2 Category distribution

The corpus is general-purpose, but personal finance is its centre of gravity. Table 1 reports the full category distribution of the primary Argentine-Spanish corpus.

Category	Calc.	Category	Calc.
Finance	462	Science	82
Everyday life	324	Family	76
Health	310	Entertainment	62
Sports	209	Automotive	59
Mathematics	157	Electronics	40
Education	128	Gardening	39
Cooking	123	Languages	33
Technology	122	Environment	30
Business	122	Taxes	18
Pets	120	Home	9
Travel	119	Weather	5
Construction	108	Games	4
Marketing	82	Astronomy	4
Total (primary corpus, AR)		2847	

Table 1. Category distribution of the primary Argentine-Spanish corpus (June 2026). Excludes the 1,277 English, Portuguese and other localized calculators.

Finance is the single largest category (462 calculators, 16% of the primary corpus); together with the closely related Business (122) and Taxes (18) categories it forms a finance-and-economics cluster of just over 600 tools. This understates the true financial footprint, because many calculators classified under Everyday life, Family, Automotive or Construction also have a monetary dimension (loan instalments, the real cost of a purchase in instalments versus cash, severance estimates, materials budgeting). The financial subset includes, among others, tools for statutory year-end bonus (aguinaldo / SAC), rent

indexation against the central-bank index, simplified-tax (monotributo) categorization and recategorization, real-wage adjustment for inflation, compound interest, and the yield of inflation-linked fixed-term deposits.

3.3 Localization, not translation

The platform is trilingual, but its financial tools are *localized*, not merely translated, because financial rules are national. A “rent-increase calculator” is not one tool in three languages; it is several distinct tools, because Argentina indexes residential rents against one published series while other jurisdictions use entirely different mechanisms. Likewise the simplified-tax tools encode Argentina's monotributo regime specifically, and the Chilean and Colombian variants incorporate their own indexation units and reference rates. Localization is therefore a substantive modelling task, not a linguistic one — and it multiplies the maintenance burden discussed next.

4. Keeping numbers honest: a data-freshness architecture

This section presents what we regard as the paper's main transferable contribution. The challenge it addresses is generic to any maintainer of high-stakes calculators; the solution is a small set of engineering patterns that any such maintainer can adopt.

4.1 The YMYL freshness problem

Search-quality and consumer-protection discourse uses the label *Your-Money-or-Your-Life* (YMYL) for content that can materially affect a person's finances, health or safety. Financial calculators are squarely YMYL: a stale tax threshold or an out-of-date index can produce a confidently-wrong number that a user acts on. Under high inflation the danger is acute, because the *half-life of correctness* is short — a parameter that was right in March may be wrong in July. Trust, once lost to a visibly wrong result, is hard to regain. The design goal is therefore not only to be correct at launch but to *fail safe* as the world moves.

4.2 Single source of truth for parameters

The first pattern is to separate volatile parameters from formulas. Domain parameters that change on an official schedule — for example, the full table of monotributo brackets and their ceilings for a given year — are held once in a dedicated data module rather than copied into each calculator that needs them. Every dependent tool imports the same canonical table, so there is no possibility of divergence between two calculators that ought to agree, and a single edit propagates everywhere at once. This converts a fan-out maintenance problem (“update forty tools”) into a point edit (“update one table”).

4.3 Automated validation and drift detection

The second pattern is to make staleness and implausibility *visible automatically* rather than relying on manual vigilance across thousands of files. The platform runs three validation passes as part of its build pipeline. A **freshness check** flags any data-driven parameter whose last-verified date exceeds a configured threshold, so that silent decay surfaces as a build warning rather than a user-facing error. A **sanity check** enforces plausibility bounds and detects suspicious drift — a value that has moved more than expected, or that disagrees with a second reference — catching both fat-finger edits and upstream data problems. An **update-validation** pass checks proposed parameter changes before they are merged. Crucially, the platform distinguishes the date a figure *refers to* from the date it was *last verified*, so that a recently-touched file is not mistaken for a recently-checked one.

4.4 Source-anchored computation: worked examples

The third pattern is to anchor each high-stakes tool to a named official source and, where possible, to refresh it programmatically.

- **Rent indexation.** Argentina's residential-rent index is published by the central bank. The platform fetches the series through a dedicated updater so that the rent-increase calculator computes the contractual adjustment from the current official index rather than a hand-copied number.
- **Simplified-tax (monotributo) categorization.** The full bracket table for the year lives in a single parameter module sourced from the tax authority; the categorization and recategorization tools read from it, so a mid-year update to the ceilings is reflected everywhere at once.
- **Statutory year-end bonus (aguinaldo / SAC).** The tool encodes the statutory formula — half of the highest monthly salary in the semester, prorated by months worked — a rule that is stable in form even as wages change, illustrating that not every parameter is volatile and that the architecture must tell the two kinds apart.
- **Real-wage adjustment.** Nominal salaries are deflated using the official consumer-price index, letting a user see whether a raise kept pace with inflation in real terms.

Each tool displays its formula and cites its source, so the computation is auditable rather than a black box — a property we return to in Section 5.

5. Design for low cognitive cost

If the goal is to reduce the *execution* cost of a financial decision, the interface must add as little extraneous load as possible. Several design choices follow from this. Each tool asks for the minimum necessary inputs and requires an explicit “Calculate” action, so that a result is never produced before the user has finished entering their situation. Inputs use plain language and sensible defaults; large monetary figures are grouped with thousands separators to reduce transcription error. The result is accompanied by the formula used and the source of any official parameter, which serves both transparency and education: a user who wants only the number gets it, while a user who wants to understand can see the working. Every tool carries a frequently-asked-questions block of at least seven entries addressing the common edge cases, and many present the result graphically. Because the pages are static and edge-served, they remain usable on low-end phones and constrained connections. None of these choices is individually novel; their combination operationalizes the principle that a calculator should lower, not raise, the cognitive cost of acting correctly.

6. Discussion

6.1 Calculators as executable rules of thumb

Section 2.3 cited evidence that simple heuristics can beat comprehensive instruction (Drexler et al., 2014). A maintained calculator is, in effect, a rule of thumb made *executable and auditable*: the correct rule is encoded once by a maintainer who can afford to get it exactly right, and is then delivered to many users at the moment of need without requiring each of them to internalize it first. This division of labour — expertise embedded in the tool, judgement retained by the user — is precisely what just-in-time financial support should look like, and it is complementary to, not a replacement for, formal education.

6.2 Equity and access

Free, account-less, low-bandwidth tools have a distributional argument in their favour. The populations most exposed to the costs of financial mistakes — lower-income households, the underbanked, informal and self-employed workers navigating the simplified-tax regime — are also those least able to pay for advice. A tool that runs on a cheap phone over a weak connection, in the user's own language and localized to their own country's rules, lowers the barrier to a correct calculation precisely where that barrier bites hardest.

6.3 Limitations

Several limitations bound the claims of this paper. First and most important, we present **no causal evidence** that the platform improves financial knowledge or decisions; establishing that requires the experiments outlined in Section 7, and the present paper is descriptive. Second, the corpus metrics are self-reported by its operator and the quality of 4,124 tools is necessarily heterogeneous; the freshness architecture mitigates but does not eliminate the risk of a stale or mis-specified tool. Third, correctness depends on the availability and accuracy of upstream official sources, which the platform does not control. Fourth, there is a **selection effect**: users who actively seek a calculator are already motivated, so any observed association with better outcomes need not generalize to the broader population that financial education aims to reach. Fifth, the author is the platform's founder and maintainer, a positionality stated plainly below.

7. Conclusion and research agenda

We have described a large, free, multilingual corpus of computational calculators and argued that, in a high-inflation economy, the decisive feature of such tools is not their initial construction but the architecture that keeps them correct as official parameters churn. The single-source-of-truth, automated-validation and source-anchoring patterns documented here are offered as a reusable template for any maintainer of high-stakes calculators.

The clear next step is evaluation. A credible agenda would (i) field randomized or quasi-experimental tests of whether using a specific calculator improves the accuracy of users' subsequent financial judgements; (ii) measure decision quality, not just self-reported confidence; (iii) track knowledge retention after tool use, to see whether executable rules of thumb leave any durable understanding behind; (iv) publish an open dataset of the official parameters and their revision history, which has research value independent of the tools; and (v) define and report an explicit freshness service-level objective, turning “kept current” from a claim into a measurable property. We invite collaboration from researchers in financial education, development economics and human-computer interaction on each of these fronts.

Data and code availability

The platform is publicly accessible at <https://hacecuentas.com>. Category hubs (for example, <https://hacecuentas.com/categoria/finanzas>) and individual tools — such as the rent-indexation, simplified-tax-categorization and statutory-bonus calculators — are openly usable without registration. Aggregate corpus statistics reported here were computed directly from the platform's content repository in June 2026. The author welcomes inquiries from researchers wishing to access parameter histories or corpus metadata for replication.

Conflict of interest and positionality

The author is the founder and maintainer of hacecuentas.com, which is a commercially-operated website funded primarily through organic search traffic. The platform carried no active monetization at the time of writing. This paper received no external funding. The author's dual role as operator and analyst is the reason the paper is framed as a descriptive and design contribution rather than an impact evaluation, and is the motivation for the independent-evaluation agenda in Section 7.

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Suggested citation

Rodríguez, M. (2026). *Open Computational Tools for Financial Literacy in High-Inflation Economies: The Case of hacecuentas.com*. Working paper. Available at <https://hacecuentas.com>